Angel Cruz

San José, CA

949-299-6284 | angel.l.cruz@sjsu.edu | linkedin.com/in/angel-cruz-gtur | github.com/angelcruzgtur

Education

San José State University — **Expected December 2027**
Bachelor of Science in Computer Science — *San José, CA*

Experience

AgileRadio — **June 2025 – Oct. 2025**
Software Engineering Intern | Python, Bash, NumPy, pandas, Jenkins — *Schenectady, NY*

- Engineered a Jenkins-based automated testing pipeline in Python and Bash, reducing manual testing workload by 40% and increasing system reliability across 12 test suites.
- Analyzed communication protocol behavior using NumPy and pandas to pinpoint performance bottlenecks, expanding test coverage by 35% and cutting defect escape rate by 28%.
- Produced technical documentation and validation reports adopted by cross-functional engineering teams, standardizing knowledge transfer across 3 departments.

Awards & Recognitions

2023 National E-Commerce Website Competition: Placed in the top 10% of teams nationally, competing against university-level participants across the United States.

CSUF Raise Program (2025): Selected as one of 20 participants from a competitive applicant pool for a STEM research initiative focused on data-driven application development.

Projects

Forge | *Python, PyTorch, PyTorch Geometric, NetworkX, Claude API* — **Jan. 2026 – May 2026**

- Built a reference-free DNA analysis pipeline that detects cancer markers and antibiotic resistance directly from raw sequencing reads, eliminating the need for a reference genome.
- Modeled sequencing reads as De Bruijn graphs and trained graph neural networks using PyTorch Geometric, achieving 91% classification accuracy on held-out test sets.
- Replaced brute-force grid search with an LLM-driven parameter optimizer, cutting tuning time by 60%, and exposed a natural language interface for non-expert graph querying.

Mini Me AI | *C++, Python, Ollama, Piper, Vosk* — **Jan. 2026 – Apr. 2026**

- Engineered a locally hosted personality-driven AI assistant with real-time speech synthesis and voice interaction, enabling fully offline conversational AI with no cloud dependency.
- Integrated Ollama for on-device LLM inference and Piper/Vosk for low-latency TTS and ASR, achieving sub-200ms end-to-end response latency on constrained hardware.
- Optimized memory allocation for embedded GPU environments, reducing peak VRAM usage by 38% and improving response consistency under sustained load.

WasteLess AI | *Next.js, React, TypeScript, Supabase, Leaflet, NVIDIA Nemotron* — **Mar. 2026**

- Built a food redistribution platform connecting restaurants with nearby shelters via a live interactive map, enabling real-time donation coordination across 30+ locations.
- Implemented a real-time availability and pickup filtering system with Leaflet and Supabase, reducing time-to-match between donors and recipients to under 3 seconds.
- Integrated NVIDIA Nemotron APIs to generate AI-assisted distribution recommendations, improving match relevance by 42% over a baseline filter-only approach.

Activities

Spartan Racing FSAE — **Jan. 2025 – Present**
Software Engineering Member | C, BUSMASTER, TASKING, PCAN — *San José, CA*

- Developed and maintained onboard vehicle data systems, contributing to measurably faster race-day diagnostics and fewer system failures during competition.
- Collaborated with mechanical and electrical sub-teams on real-time troubleshooting during testing sessions, directly resolving 8 critical issues before race events.
- Produced competition media for sponsor outreach and team marketing, supporting visibility efforts for 6 active sponsors.

Technical Skills

Languages: C++, Java, Python, JavaScript, SQL, HTML/CSS
Frameworks: React, Next.js, FastAPI, Streamlit
Tools: Git, GitHub, Jenkins, Postman, Excel, VS Code
Technologies: Linux, Supabase, REST APIs, Ollama, Piper, Vosk
Spoken Languages: Spanish (Fluent), English (Fluent)